run_sjsdm_common_regularization_sensitivity.R

October 22, 2026

Contents

run_sjsdm_common_regularization_sensitivity 1

Index **3**

`run_sjsdm_common_regularization_sensitivity`
Run Common-Regularization sjSDM Sensitivity Models

Description

Reads prepared representative-model inputs, refits each model with a common spatial regularization artifact, and returns decomposition and provenance.

Usage

```
run_sjsdm_common_regularization_sensitivity(
  data_model_index = NULL,
  data_artifacts = NULL,
  read_target_function = targets::tar_read_raw,
  fit_function = NULL,
  standard_error_function = compute_jsdm_se,
  anova_function = compute_jsdm_variance_partition_with_progress,
  extract_function = extract_jsdm_variance_fractions,
  path_checkpoint = NULL,
  verbose = TRUE
)
```

Arguments

`data_model_index`
: Model index with model_id, tier_id, scale_id, resolution_id, and store_path.

`data_artifacts`
: Common artifacts from [build_sjsdm_common_regularization_artifacts()].

`read_target_function`
: Injectable target reader. Defaults to [targets::tar_read_raw()].

`fit_function` Injectable high-level model-fitting function. It receives prepared input, formula, fitting configuration, and one regularization artifact.

`standard_error_function` Injectable standard-error function. Defaults to [compute_jsdm_se()].

`anova_function` Injectable decomposition function. Defaults to [compute_jsdm_variance_partition_with_progress()].

`extract_function` Injectable ANOVA extractor. Defaults to [extract_jsdm_variance_fractions()].

`path_checkpoint` Optional root for per-model, per-phase checkpoints.

`verbose` Logical; print representative and phase progress.

Details

Checkpoints are independent of publication timestamps. Fitting inputs, formula text, regularization parameters, budgets, seed, and backend code identify reusable phases. Changing only the ANOVA sample count reuses the fit and standard errors. Decompositions resume at phase boundaries, not at an interrupted likelihood taxon. The formula is evaluated against the prepared model inputs; its incidental calling environment is not hashed.

Value

Named list containing fitted models, ANOVA objects, model-level provenance, and a long decomposition table. Individual failures are retained with an explicit status and do not discard successful representative models.

Examples

```
## Not run:
run_sjsdm_common_regularization_sensitivity(
  data_model_index = data_model_index,
  data_artifacts = data_artifacts,
  path_checkpoint = "Data/Temp/sensitivity_checkpoints"
)

## End(Not run)
```

Index

run_sjsdm_common_regularization_sensitivity,
        1